HEMIANOPIA_TAMANNA

TrialLogger.cs is a recorder. While it's running, it writes one line to a spreadsheet file for every frame of the headset display (about 72 per second), noting what the eyes, head and experiment were doing at that moment.

What's in each line

- **Timing:** when the frame happened, and whether the eye tracker gave a fresh reading this frame. The tracker is slower than the display, so some frames reuse the old reading.
- **Where the eyes were pointing:** the raw tracker reading, and the version after the calibration correction, both in degrees left/right and up/down.
- **Whether the tracker was working:** raw_valid and since_new_ms, which catch blinks and dropouts.
- **Head position and rotation, and how fast the head was turning:** this matters because Russell's tests found head turns cause about 1° of error.
- **Fixation info:** the 3D spot being looked at, and whether the person was holding a steady gaze.
- **Filter state:** whether the hemianopia mask was on, and which side was blind.
- **Experiment info:** trial number, condition (like "no cue" or "smart cue"), where the target was, whether the person could see it, and a label for one-off moments like a button press.

Movingblock.cs

How to run it

- **Keyboard:** 1 = square, 2 = X, 3 = saccade steps, 4 = midline crossing, 0 or Esc = abort.
- **Controllers:** push the right thumbstick **right** to pick a mode, **down** to start or stop it.

Each run starts the logger on its own and stops it when the path finishes. Keep your head still and follow the block.

What each mode does

Mode	What happens	Use it for
Square	Glides around a ±10° square at 15°/s, 2 laps	Horizontal and vertical pursuit, lag estimate

X	Glides along both diagonals through the center	Diagonal pursuit
Saccade steps	Jumps between fixed pairs (horizontal, vertical, two diagonals), 1.5 s at each point	Landing accuracy
Midline crossing	You fixate a cyan center dot while the block slides slowly left to right and back	The first half of the mask-leak check

The log marks events (run start and end, laps, each jump, the moment the block crosses the midline), so your analysis can cut the data into pieces without guessing.

Things I decided that you should know about

- **The block moves with your head.** It sits at fixed angles in the head's frame, like the calibration dots, which removes head motion from these tests. That's also why you should keep your head still.
- **An invisible depth wall appears during runs.** The filter works out each eye's boundary from the 3D spot you're looking at, found by raycasting. With nothing to hit, it falls back to 100 m, which shifts the boundary by close to a degree at 2 m. So during a run I add an invisible wall at the task distance, and your fixation depth is always right. There's a checkbox (addDepthPlane) to turn it off if Russell prefers otherwise.
- **tgt_visible in the log is an estimate.** It says where the filter should be hiding the block, based on the block's angle versus your gaze angle. The real filter works per eye in 3D, so treat this as ground truth to compare against, not a perfect answer.

What to test first

1. Run mode 1 with the filter off. The block should trace a square smoothly and your gaze reticle should follow it.
2. Load the CSV and plot cor_az against time next to the block's path. They should have the same shape, with the gaze slightly behind.
3. Run mode 4 with the filter on (H key). The block should vanish as it crosses to the blind side of your gaze point. Note whether it flickers or leaks near the dot.